Baloney Detectives Progress Report

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Abstract:

Misinformation is a pressing global issue; it is increasingly easy to post false or misleading information on the internet and reach massive audiences. *Baloney Detectives* is a virtual escape room game made to mimic the psychological and emotional responses elicited by real-life interactions with digital misinformation. Made with the game engine Twine, the escape room aims to equip players with the necessary skills to identify misinformation online and become savvy, skeptical digital citizens.

Keywords:
Game design, misinformation, information literacy, educational tools, social media

Research:

We are addressing the convergent issues of misinformation and disinformation in social media through game-based learning. We are specifically addressing it by looking at the roles social media influencers play in spreading unintentionally or intentionally false information on the Internet. Our goal is to help teenagers learn techniques for identifying mis- and disinformation on social media. Our prototype is a way for them to be actively involved in “detective” work regarding health misinformation, something considered rampant in contemporary social media landscapes thanks to the ideologies spread by the brands of public figures.

According to a DataReportal survey, the 13-17 age range only makes up 6.6% of the Twitter user-base as of 2021 (Kemp). In 2022, Pew Research Center conducted a survey within the same age range and came up with 23% of teens saying they have ever used Twitter (Atske). It has been difficult to identify one number for what percentage of the userbase are teenagers due to surveys using different sampling methodologies and bin sizes. Even if only 27% of current Twitter users are teens, the digital literacy concepts we are including in our game are applicable to other social media platforms. Twitter posts and influencers are also often shared through other social media or word of mouth. Digital literacy skills such as identifying fake posts and unreliable sources, however, are very valuable for evaluating information beyond Twitter itself.

Teens are our main stakeholders, as they are the targeted demographic, and their goal is to improve in information literacy and the detection of mis- or disinformation, which is a goal we are collaborating with them to achieve. Our project is also partially aimed at schools and libraries to purchase or adopt, as they may want to integrate the game into teen programming so that they can train young people in navigating information networks in the digital age and build skills in effective information literacy.

Teens are more susceptible to influencer messaging. According to the National Association of Anorexia Nervosa and Associated Disorders, 46% of 9-11 year-olds are either often or sometimes on diets and 35-57% of adolescent girls diet, fast, take diet pills, or engage in other potentially dangerous eating habits ("Eating Disorder Statistics"). We have thus identified a need to target misinformation surrounding harmful eating habits and promotion of diet products.

Design Notes:

Observation and Framing:

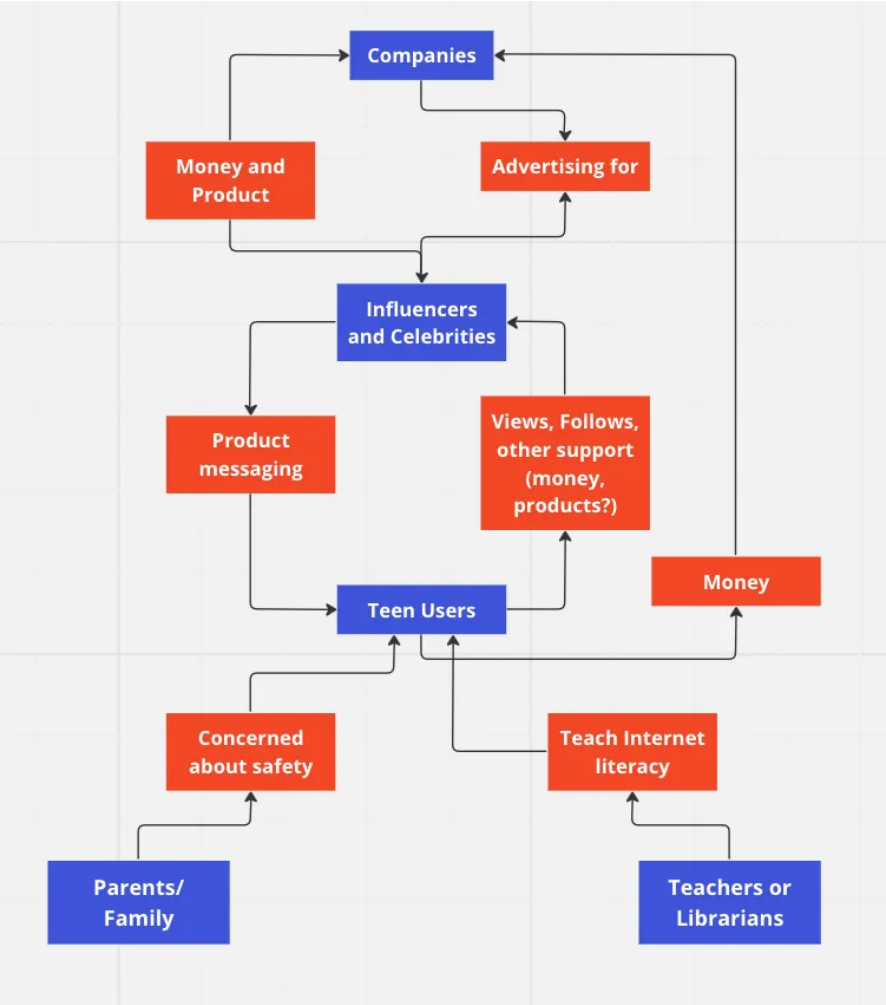
The Internet, chaotic as it is, has lots of potential for learning but also for harmful misinformation. Sharing misinformation is effortless at times, especially on platforms like Twitter. Social media, in particular, allows for the quick, easy sharing of stories and information in short form, through the interface allowing for quick scrolling of massive amounts of content. It is easy to see what friends and colleagues are interested in based on what information is shared. Influencers on social media in general have many followers to whom they can promote and circulate ideas or advertise products, using their social presence as a platform, and users are more likely to trust posts shared by them due to that large following.

Our project is an example of gamification, a design practice described by the American Library Association as "the application of game-based elements and game design techniques to non-game settings" (ALA, 2022). The usage of play-based elements in educational settings has seen a rise in the last decade, as educators have noted that bringing in styles of learning that children and teens may already be passionate about helps increase their motivation and socialization (McGonigal, 2011). Bringing gaming into education promotes curiosity, flexible thinking, and diverse styles of literacy, and our escape room's focus on those values as well as information literacy is a strong example of how gaming concepts can support solutions to real-world issues. Educational technology companies such as Loki's Loop and the Institute for the Future are using escape rooms in particular, which have seen a "meteoric" rise in the late 2010s and early 2020s, to offer a more diverse range of game options that reflect youth's main gaming interests (Coward, 2021; Hill, 2014).

Educational gaming offers an important way to develop emerging and traditional forms of "literacy" together, and the usage of *Baloney Detectives* in hypothetical library and education settings can help teens develop systems thinking and information literacy skills together, which will become increasingly important in work environments where jobs are regularly changing.

Stakeholder Diagram:

This diagram was made using the design software Figma, and offers a visual representation of interactions between different stakeholders in our game's design situation. These interactions happen in both real-world and simulated narratives around misinformation and information literacy, and helped guide our prototype.



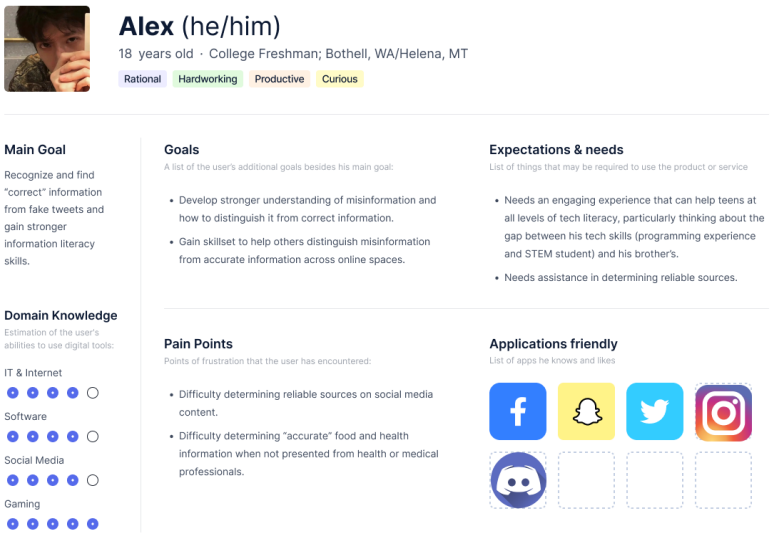
Stakeholder Interviews / Context Studies:

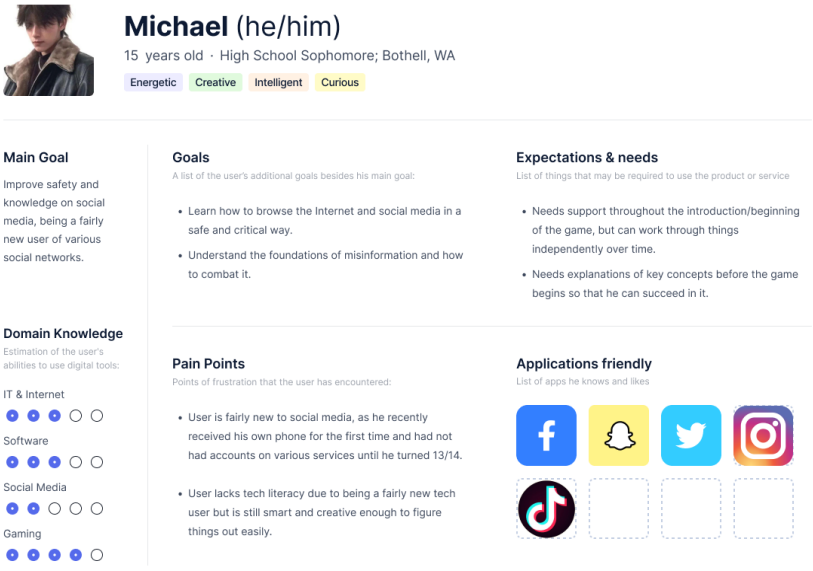
Our first interview was with Dr. Jason Yip, a professor at UW who is in charge of the Digital Youth Lab and researches game-based learning. We were recommended in the last writing assignment to reach out to an expert before interviewing teens, and Professor Yip was specifically recommended as a domain expert for both "halves" of our project (the gaming and education). Professor Yip showed us the importance of learning curves in games and how games themed after digital or information literacy in particular needs to be accessible in terms of language and design.

We luckily had two teens that were more than willing to work with us to develop the game further: Joe's teenage cousins Michael (age 15) and Alex (age 18), both of whom represent direct stakeholders at a variety of technology fluency and information literacy levels. Since they both enjoy video games and

computers, we thought that it would be useful to gain an understanding of their specific needs through the interview.

We developed user personas for the cousins on Figma, based on the interviews, that helped inform how we could tailor the project to a variety of digital and information literacy levels. They are shown below, and we also created an affinity diagram based on their interviews.





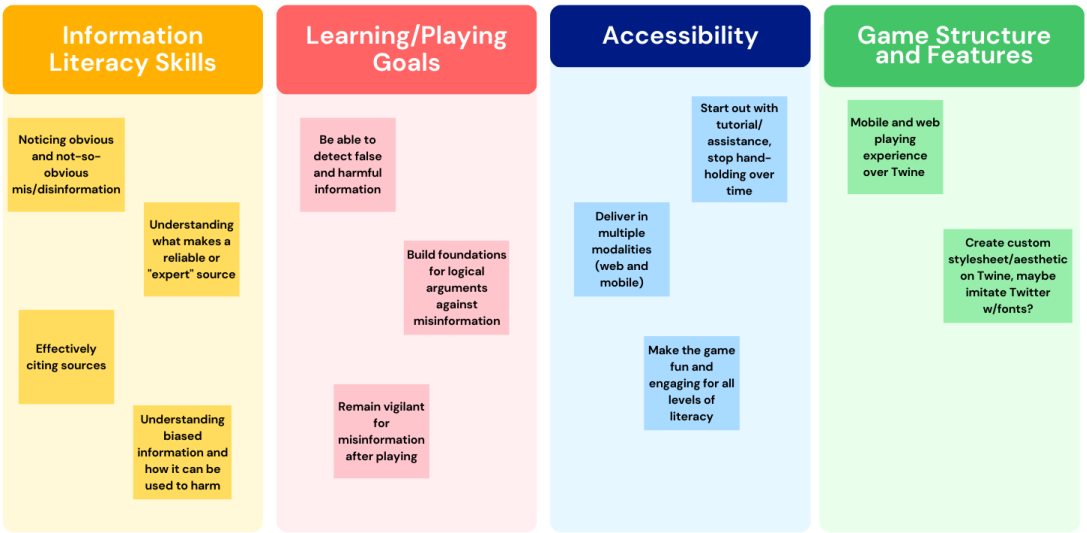
User Stories:

Scenario #1:

Ethan is a teenager who wants to become a social media influencer. He wants to learn about the consequences of spreading misinformation and disinformation on social media, and gain a better understanding of his responsibilities with his followers.

Affinity Diagram:

This affinity diagram was made on Figma and was based on insights gathered from the interviews with Joe's cousins, which suggested some features for the game and some learning and playing goals to utilize.



Prototype Progress:

We have created a low-fidelity prototype of *Baloney Detectives* over Figma. It is a basic "wireframe" display of how we eventually want the puzzles and interactions to look on the web. It contains four puzzles which go from easy to hard – we are presently working on one final level that synthesizes all the concepts discussed. This would ideally ask users to combine all the skills gathered throughout the game as a "boss" level of sorts. However, the current prototype is operational and should be more than enough to gain relevant feedback.

Our next step will be making the high-fidelity prototype, a fully operational escape room over Twine. Andrew has edited the game's story and is working on the final puzzle. Luka is using Twine's PHP scripts for the back-end's branching and puzzles, while Sabrina and Joe are working on the front-end interface and interaction using Twine'sHTML and CSS (including copying and pasting Andrew's writing). Twine games are luckily available to play over both web browsers and phones, which will make the game accessible to as many stakeholders as possible. The group members taking on programming tasks are familiar with their respective programming languages. We are presently looking for a place to host the game, with Luka recommending the game publishing website itch.io.

Evaluation:

User-centered design is crucial for designing effective educational tools

Throughout the design process, we learned that designing effective educational tools requires a user-centered approach. By understanding the needs, preferences, and motivations of the target audience, we can create tools that are engaging and effective in achieving their goals. Through user feedback, we were able to develop a game prototype that was more effective in promoting digital literacy skills among teenagers.

Collaboration is essential for addressing complex challenges

Another important insight that we gained during the design process was the importance of collaboration and interdisciplinary approaches in addressing complex challenges. Wicked problems, such as the challenge of misinformation and disinformation, require diverse perspectives and skills to address. By collaborating with teammates from different backgrounds and perspectives, we were able to develop a more comprehensive and effective approach to designing a digital literacy game.

Flexibility and adaptability are necessary for designing solutions

Lastly, we learned that designing solutions requires flexibility and adaptability. Solutions to problems are not static and must be able to evolve and adapt to the problem itself. By designing solutions that are flexible and adaptable, we were able to create a game prototype that could continue to promote digital literacy skills even as the landscape of misinformation and disinformation evolves over time.

Implications:

The project has implications for libraries in that it demonstrates how using gamification could work in terms of mis/disinformation. One of the project's goals was to enable our project to serve as a prototype. We wanted to empower teachers and librarians to utilize our game for their other goals. The prototype could help give libraries an example of how to incorporate mis/disinformation.

The project has implications for the world at large in that it demonstrates the feasibility of this idea. Teachers could construct a game like this with relatively little resources in order to help enhance mis/disinformation education. This prototype could be reconfigured in order to help with any number of subjects. We hope this project could serve a role of demonstrating the feasibility of gamification. We think this demonstration of feasibility is our major implication. The project may also demonstrate the usage of things like humor within a learning environment. We may have been able to demonstrate the lack of current mis/disinformation education among teens with additional interviews. I think that this is still a reasonable implication, but it could be more provable with additional research.

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